الجامعة المصرية اليابانية للعلوم و التكنولوجيا

エジプト日本科学技術大学

EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Experiment 04- Bipolar Junction Transistor Characteristics

Name: sohyla said Mohamed

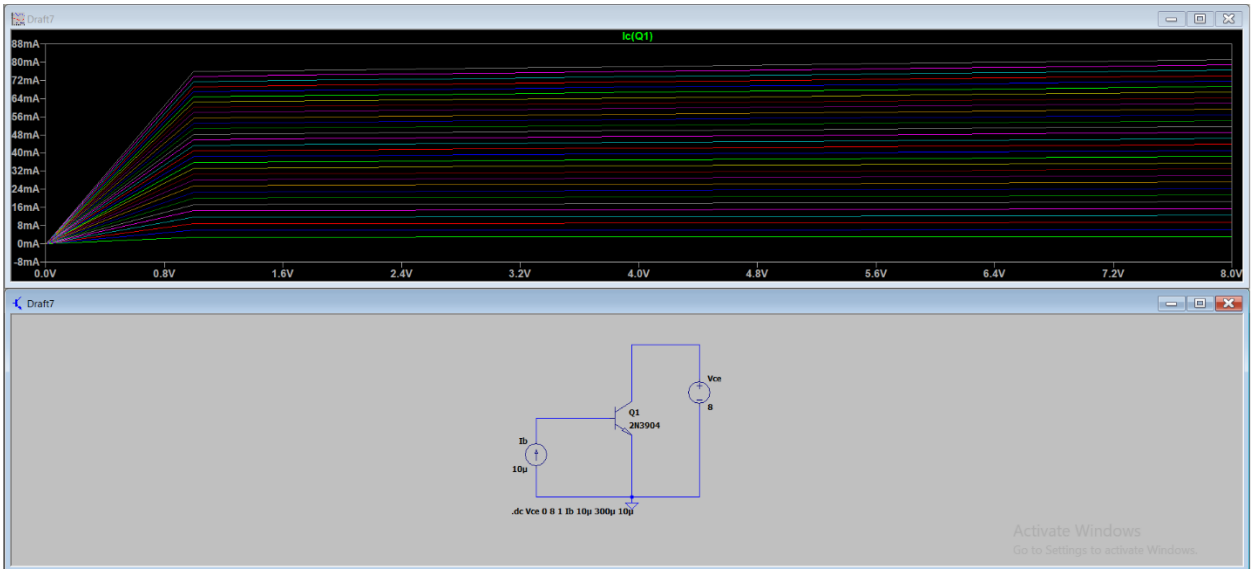
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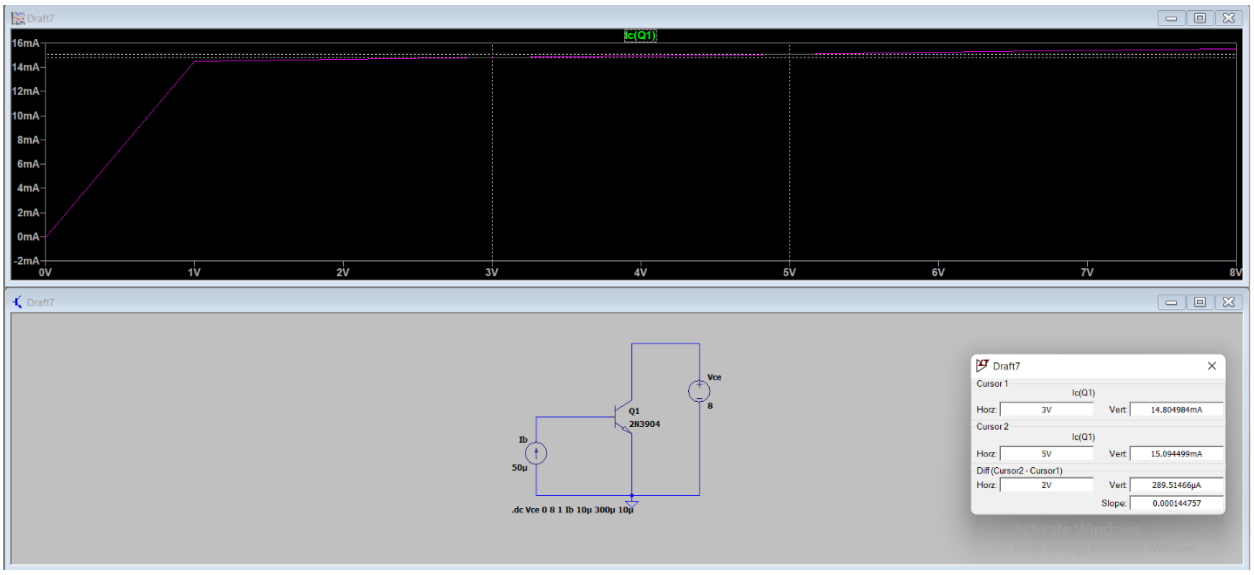
Prepared by

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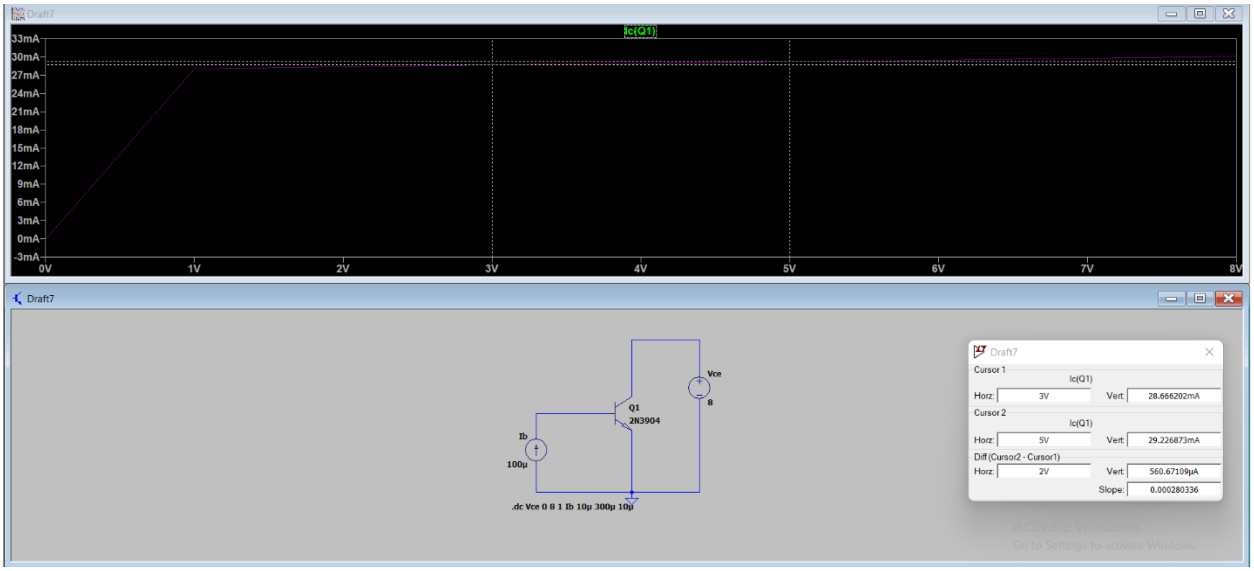
A)



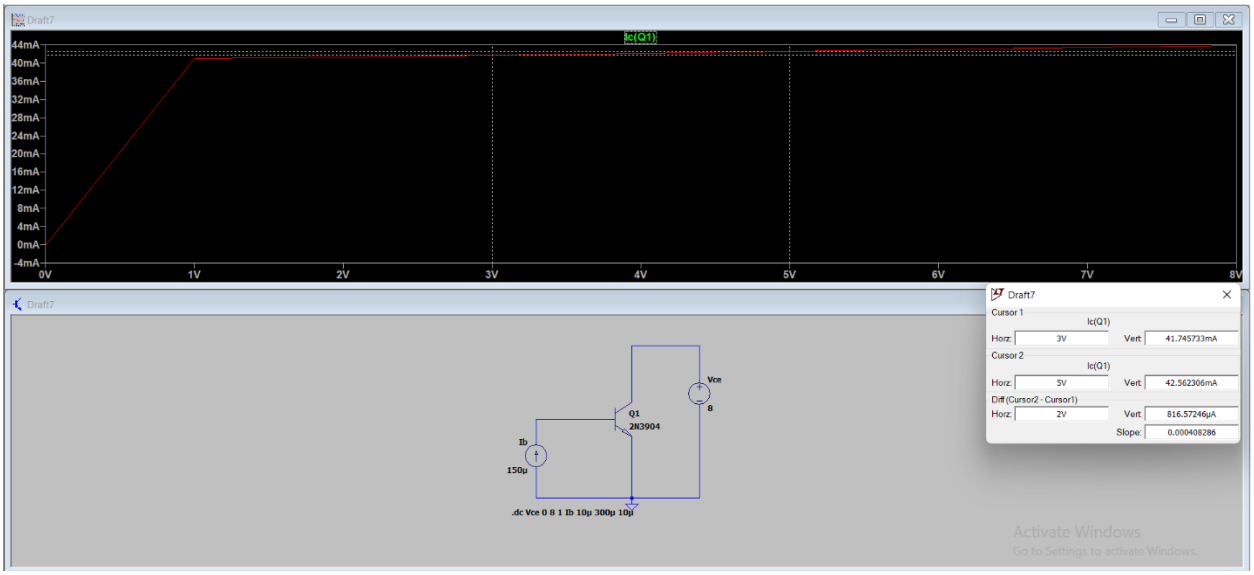
ALL Value of Ic VS Vce at all values of Ib



Value of Ic VS Vce at Ib= 50μA at 3V and 5V



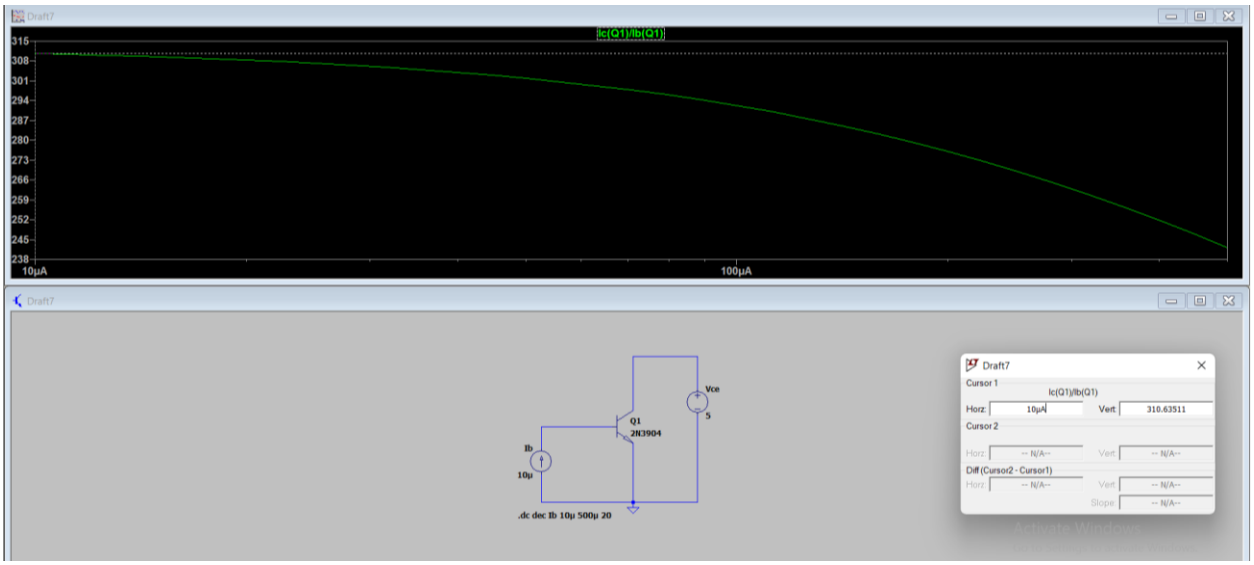
Value of Ic VS Vce at Ib= 100μA at 3V and 5V



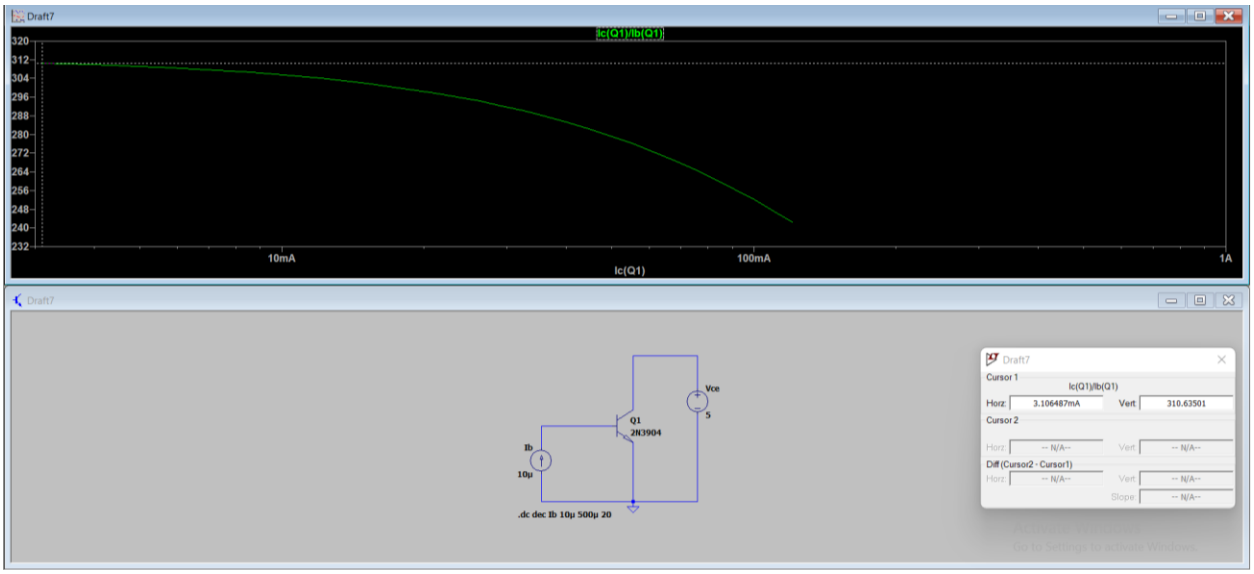
Value of Ic VS Vce at Ib= 150μA at 3V and 5V

Ib	Ic at 3V	Ic at 5V	β at 3V	β at 5V	α at 3V	α at 5V
50uA	14.8 mA	15.1 mA	296	302	0.996633	0.9966997
100uA	28.7 mA	29.2 mA	287	292	0.996528	0.9965870
150uA	41.7 mA	42.6 mA	278	284	0.996416	0.9964912

B)



Graph between Ic(Q1)/Ib(Q1) vs Ib



Graph between Ic(Q1)/Ib(Q1) vs Ic(Q1)

Maximum (β) = 310 at Ic=3.106mA and Ib=10uA.

α=310/(310+1)=0.9967845659